Digital Twin of an Ultrasonic Fermentation System

A Proof of Concept

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Context: Beer Fermentation & Ultrasound

- Beer fermentation is a **slow, resource-intensive** industrial process: experiments take days and are costly to repeat
- **Ultrasound technology** (20–50 kHz, low power) is a promising tool to accelerate yeast activity

Yeastime estimates: a 30% reduction in fermentation time in craft breweries could lead to cost savings of approximately **10–15%**

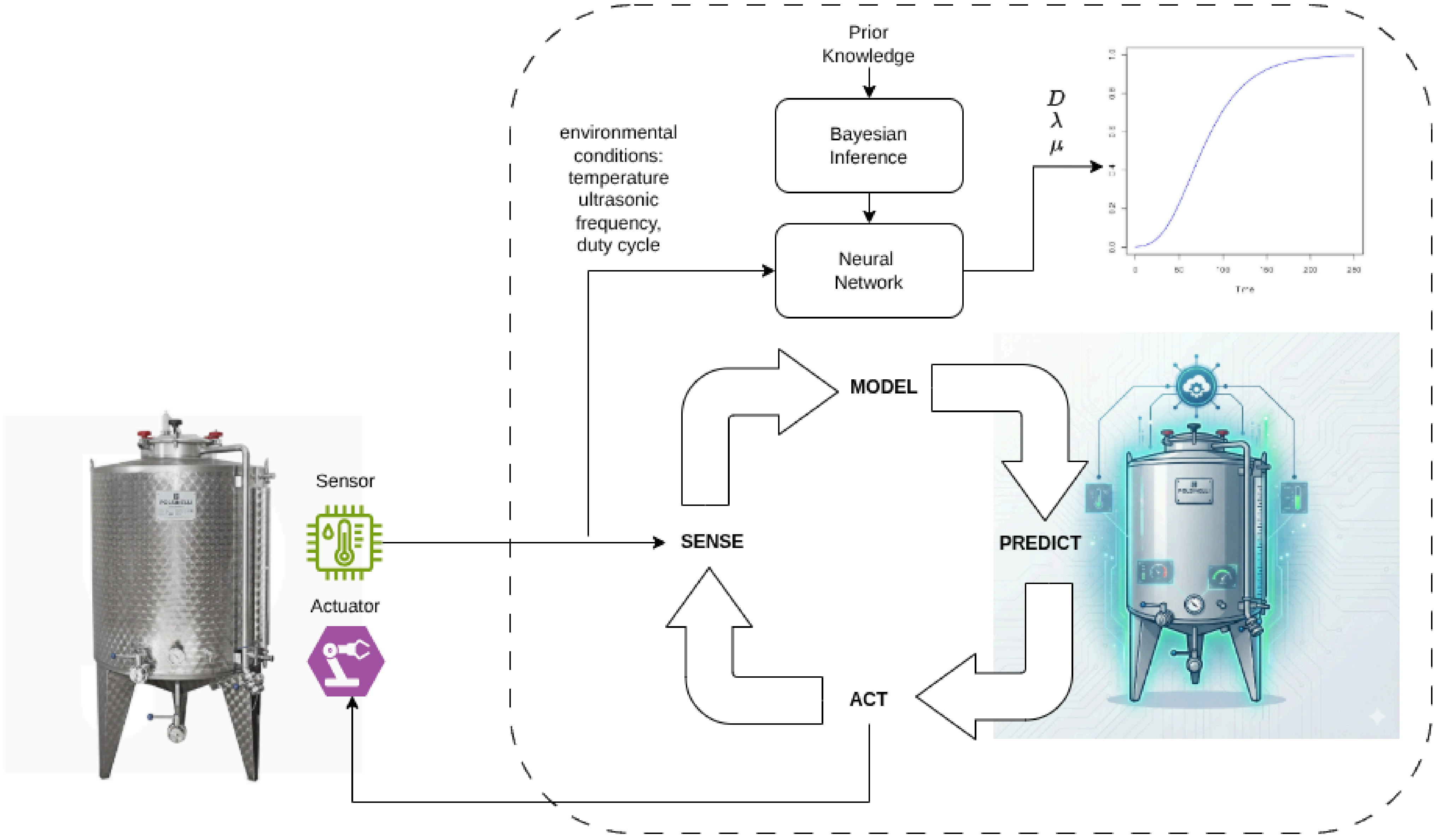
- Key effects: improved mass transfer, increased cell permeability, reduced inhibitory metabolites
- This work presents the **first digital twin tailored to an ultrasound-treated fermentation system**

What is a Digital Twin?

A **digital twin** is a virtual representation of a physical system that stays continuously synchronised through real-time data.

Key Benefit

Physical fermentation trials are **slow and expensive**, the digital twin complements them with **rapid simulation (growth model) and intelligent actuation**.



Physical Fermentator

Digtal Twin of the Fermentator

The Dataset

Biological datasets are limited by **slow, manual sampling** each growth curve takes days

⚠ Data Scarcity Challenge

~100 data points: approximately **100× smaller** than the reference study [Palacios et al., 2014]

Classical training is ineffective: high risk of overfitting.

The Dataset

Feature	Description
Ultrasound duty cycle	Fraction of time transducer is active
Irradiation frequency	Ultrasonic frequency applied (normalised)
Temperature	Medium temperature (normalised)
Initial / final Optical Density	Yeast density measured at 600 nm

> Points sharing the same duty cycle, frequency, and initial density belong to the **same biological growth curve**: treated as a group in cross-validation.

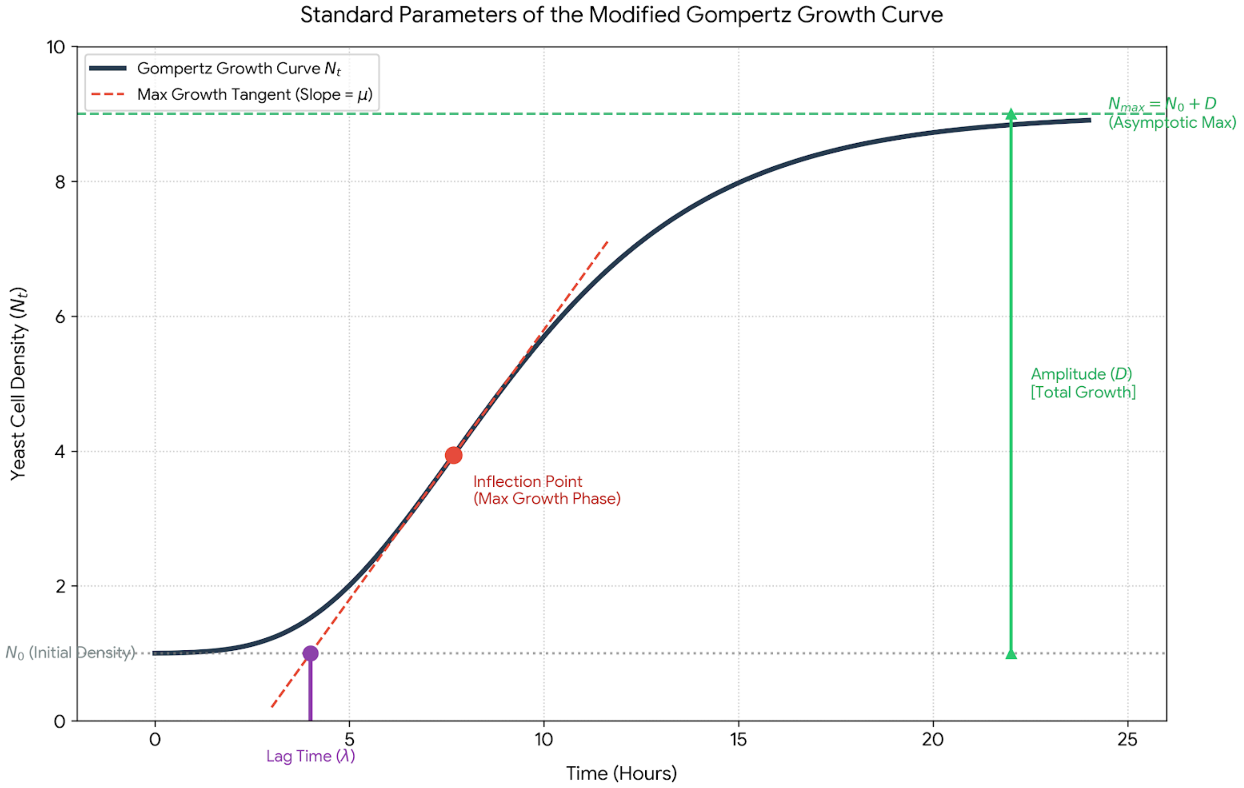
Gompertz Growth Model

- **D** = total growth amplitude
- **μ** = max growth rate
- **λ** = lag time

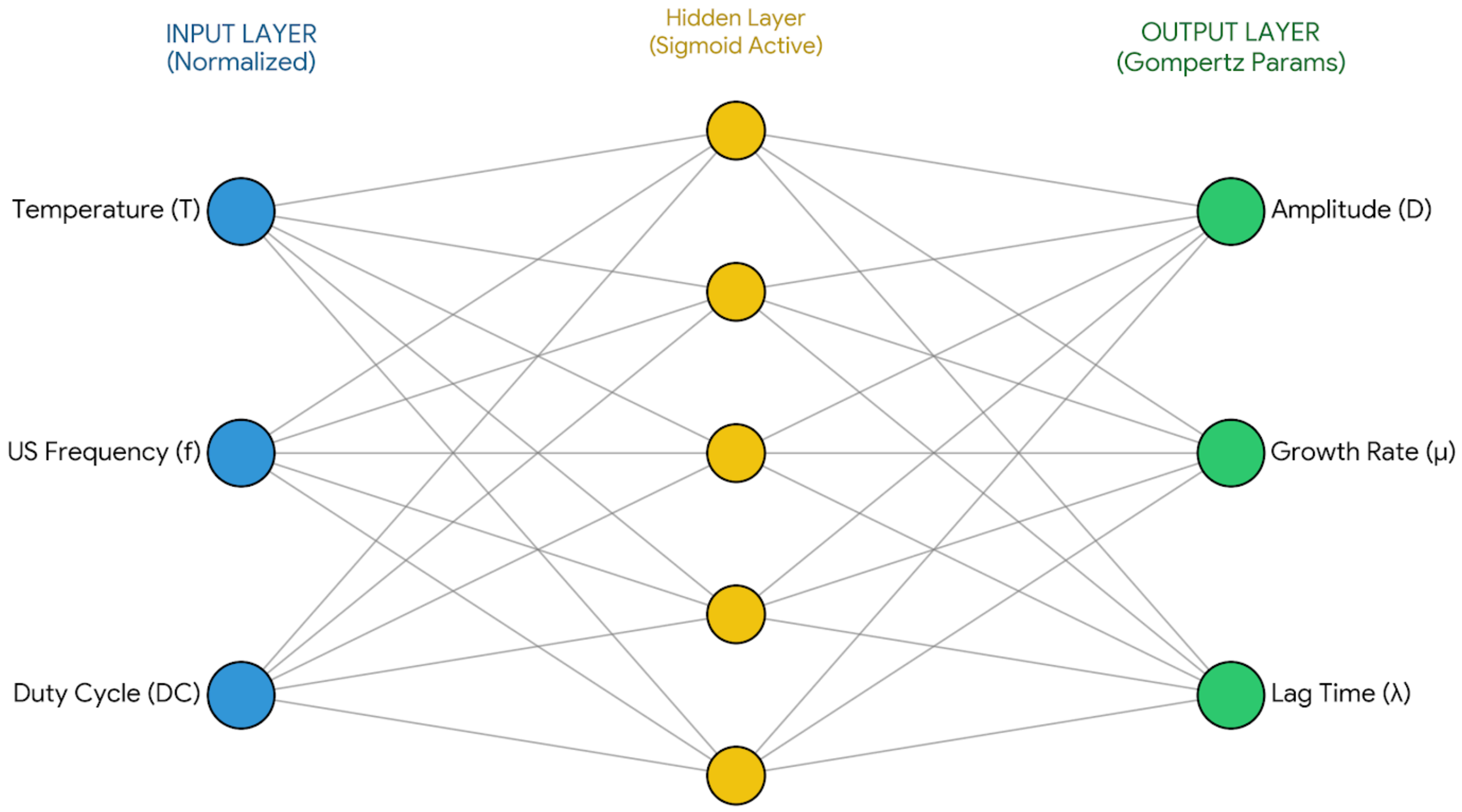
Predicted by a Bayesian Neural Network.

Yeast Density at time t:

$$E[N_t \mid N_0, D, \mu, \lambda] = N_0 + D \cdot \exp\left(-\exp\left(1 + \frac{\mu e(\lambda - t)}{D}\right)\right)$$



Goal


INPUT LAYER
(Normalized)
Hidden Layer
(Sigmoid Active)
OUTPUT LAYER
(Gompertz Params)
Temperature (T)
US Frequency (f)
Duty Cycle (DC)
Amplitude (D)
Growth Rate (μ)
Lag Time (λ)

Priors

We have "clear", physical intuition for the outputs (D, μ, λ), but we have zero biological intuition for what individual neural network weights (w) should look like.

Translate a Prior on the Output Space (D, μ, λ) into a Prior on the Parameter Space (w) via the Likelihood function and MCMC sampling.

Detour

The posterior probability of a set of weights $\mathbf{w}$ given the observed fermentation data y is:

$$p(\mathbf{w} \mid y) \propto p(y \mid \mathbf{w}) \cdot p(\mathbf{w})$$

- $p(y \mid \mathbf{w})$ is the Likelihood: How well do the weights explain the raw sensor data?
- $p(\mathbf{w})$ is the Weight Prior

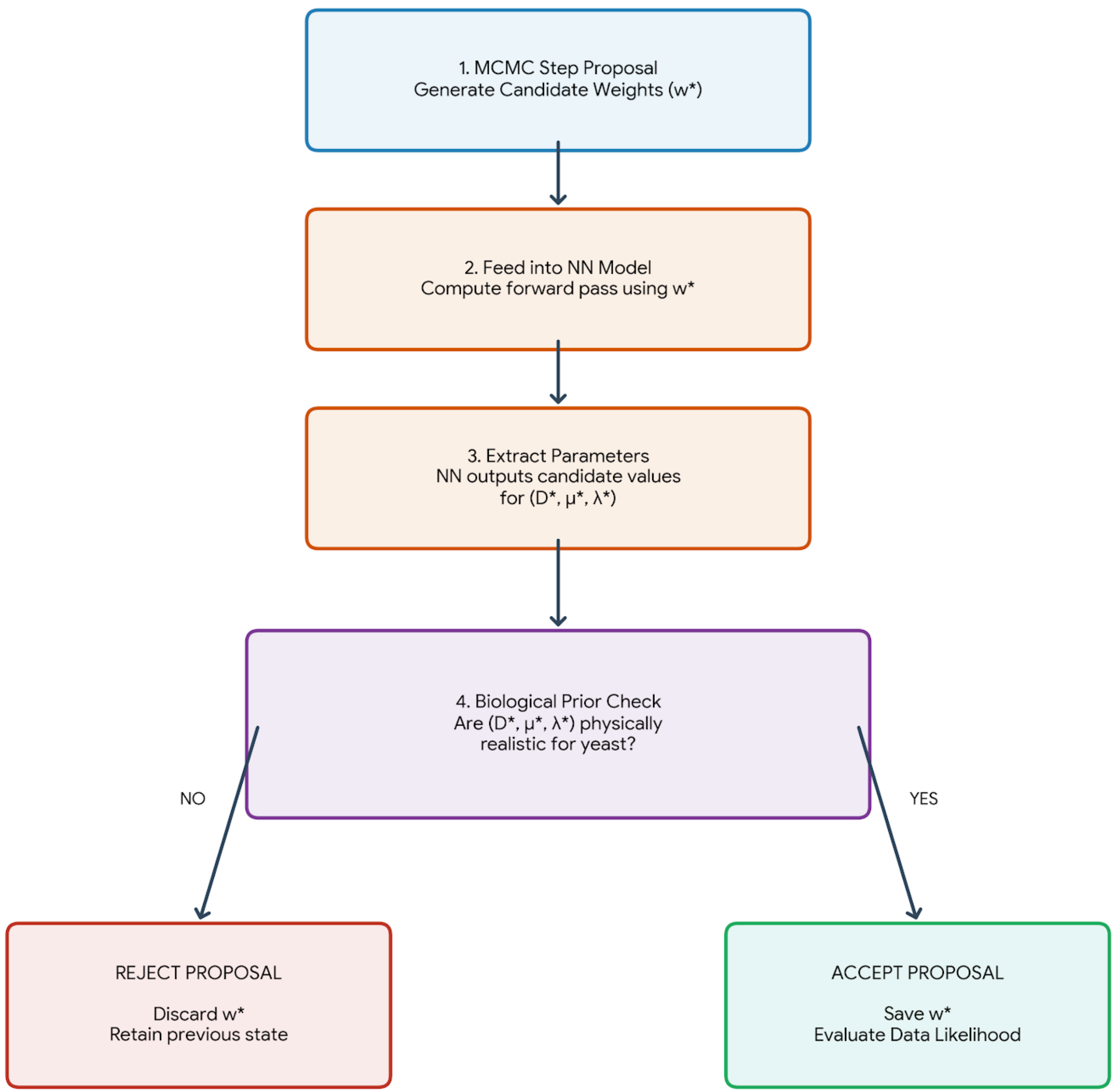
If a configuration of weights $\mathbf{w}^*$ causes the impossible value for D, μ, or $\lambda \implies P(\mathbf{w}^*) = 0$

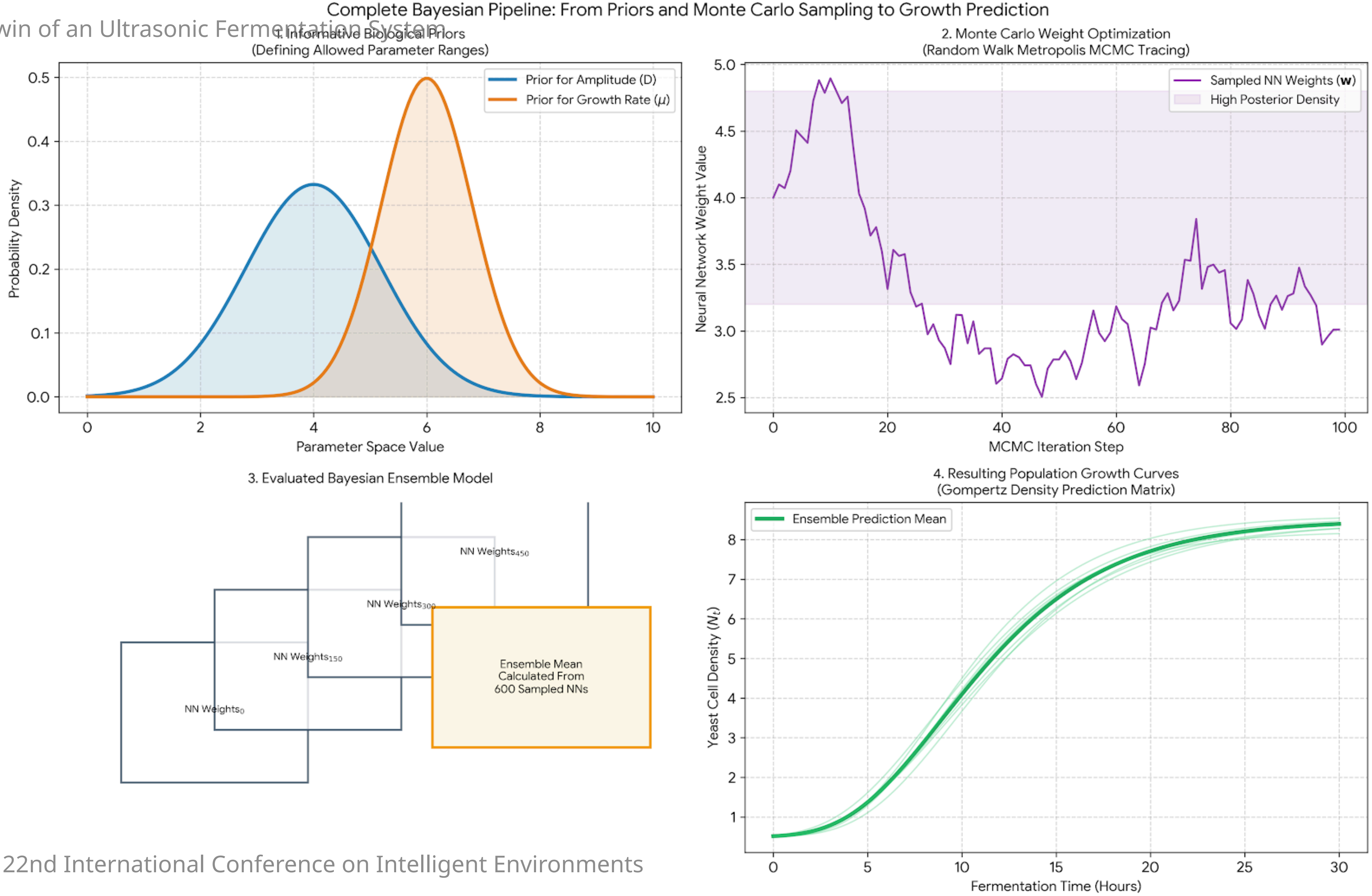
The biological output limits act as a filter that shapes the allowed weight space.

Monte Carlo (MCMC) Sampling

Because we can't analytically calculate how to constrain the weights to get the right outputs, we let the Random Walk Metropolis algorithm discover the valid weight spaces through exploration.

MCMC Induced Weight Selection via Biological Output Filtering




Complete Bayesian Pipeline: From Priors and Monte Carlo Sampling to Growth Prediction
1. Informative Biological Priors
(Defining Allowed Parameter Ranges)
Prior for Amplitude (D)
Prior for Growth Rate (μ)
Probability Density
Parameter Space Value
2. Monte Carlo Weight Optimization
(Random Walk Metropolis MCMC Tracing)
Sampled NN Weights (**w**)
High Posterior Density
Neural Network Weight Value
MCMC Iteration Step
3. Evaluated Bayesian Ensemble Model
NN Weights$_{450}$
NN Weights$_{300}$
NN Weights$_{150}$
NN Weights$_{0}$
Ensemble Mean
Calculated From
600 Sampled NNs
4. Resulting Population Growth Curves
(Gompertz Density Prediction Matrix)
Ensemble Prediction Mean
Yeast Cell Density (N_t)
Fermentation Time (Hours)

Bayesian Inference Pipeline

Given the **limited dataset**, classical training is ineffective. We adopt [Palacios et al., 2014]:

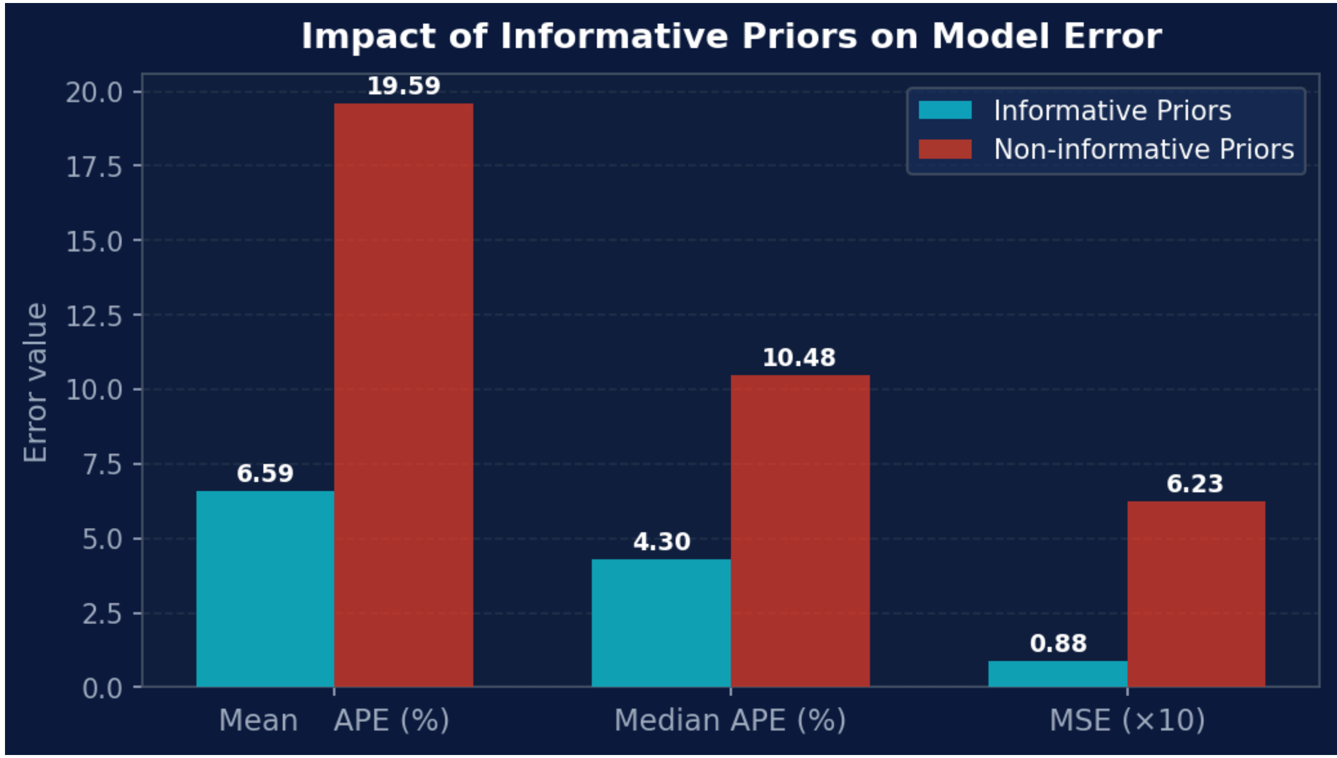
1. **Prior assignment**: probability distributions encode biological domain knowledge
2. **Random Walk Metropolis**: explores the posterior to find valid weight configurations; Metropolis-Hastings criterion escapes local maxima
3. **Network ensemble**: each accepted MCMC state instantiates a distinct NN outputting D, μ, λ
4. **Final prediction**: parameters averaged across all sampled networks → Gompertz curve

For **small datasets**, highly informative priors are essential. For larger datasets, weakly informative priors such as $\mathcal{N}(0, 100)$ are sufficient (as also recommended in [Palacios et al.]).

Informative Priors Matter

Removing informative priors causes a **~7× increase in MSE**:

- Mean Absolute Percentage Error (Mean APE)
- Median Absolute Percentage Error (Median APE)
- Mean Square Error (MSE)



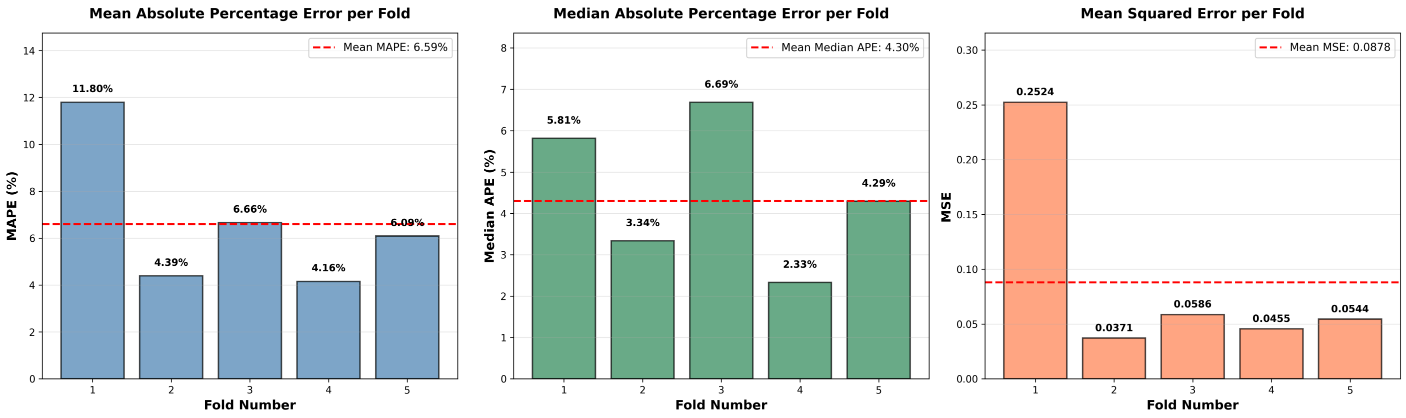
⚠ Takeaway

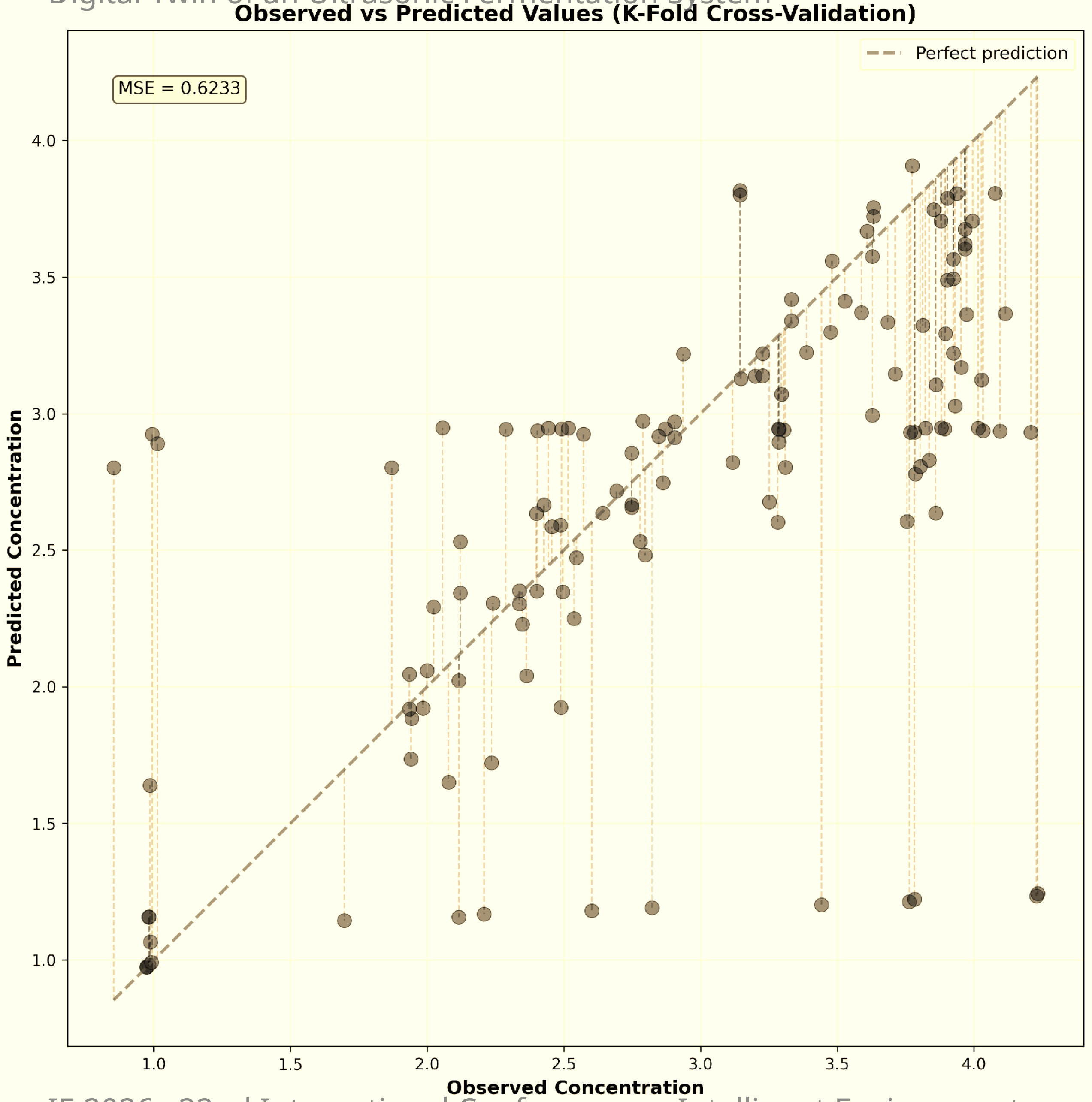
In fermentation, experiments are too slow to generate large datasets. **Biological prior knowledge is the only practical regulariser**.

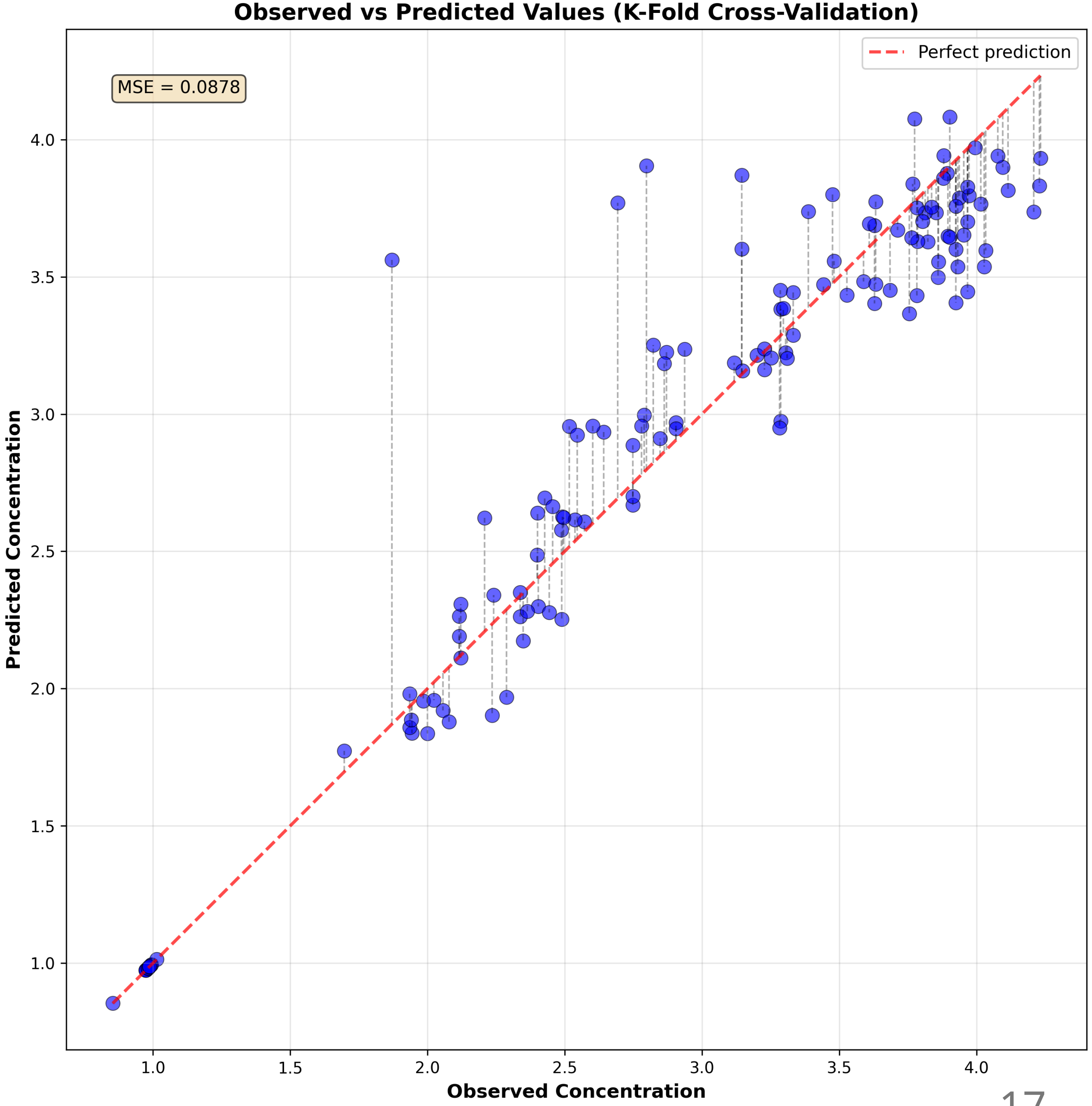
Model Performance: 5-Fold Cross-Validation

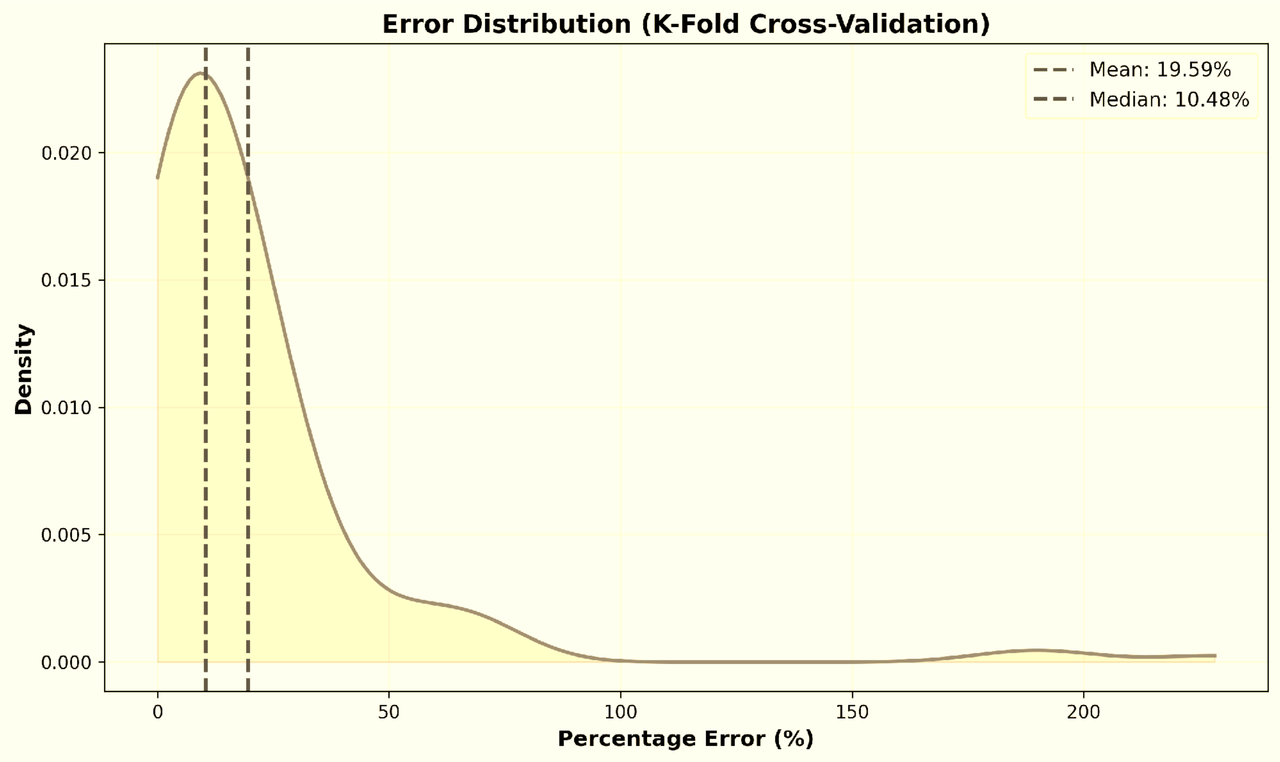
MSE is ~88× higher than [Palacios et al.]

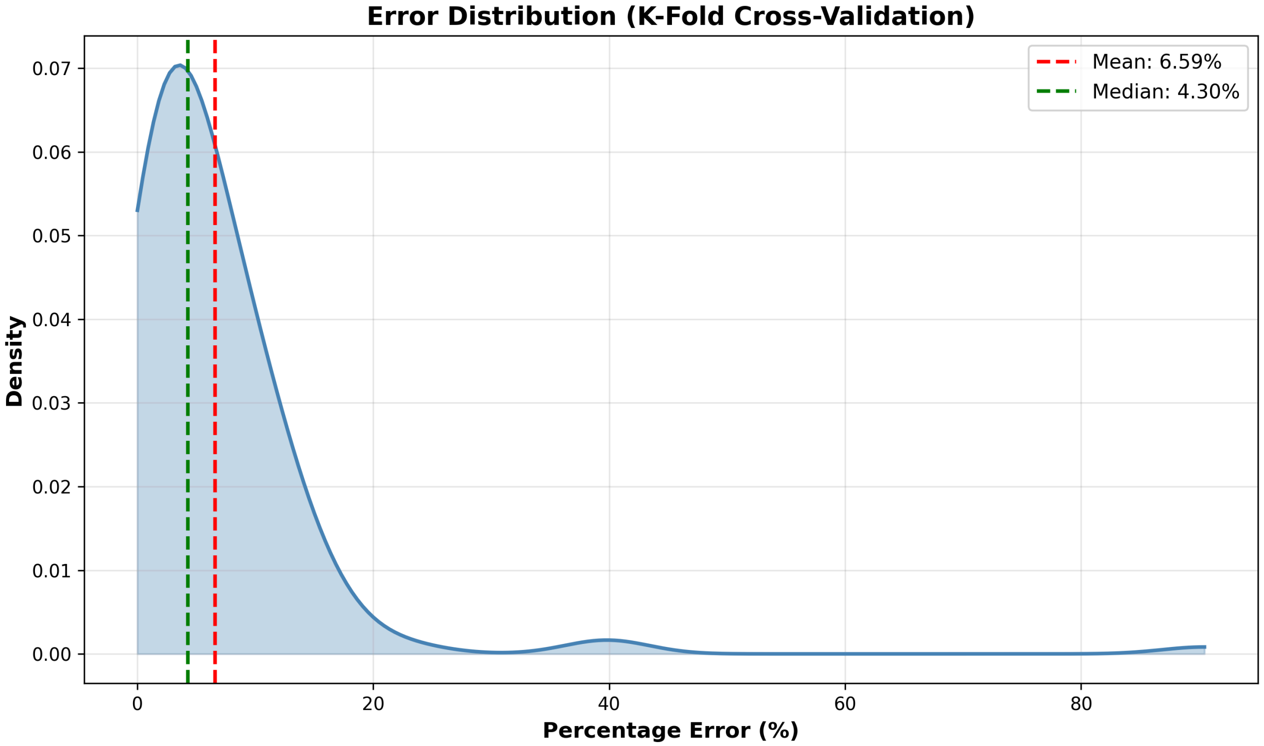
- dataset size (~100× smaller)
- single MCMC chain
- optical density (OD) measurement scale differences.




Observed vs Predicted Values (K-Fold Cross-Validation)
Perfect prediction
MSE = 0.6233
Predicted Concentration
Observed Concentration


Observed vs Predicted Values (K-Fold Cross-Validation)
Perfect prediction
MSE = 0.0878
Predicted Concentration
Observed Concentration

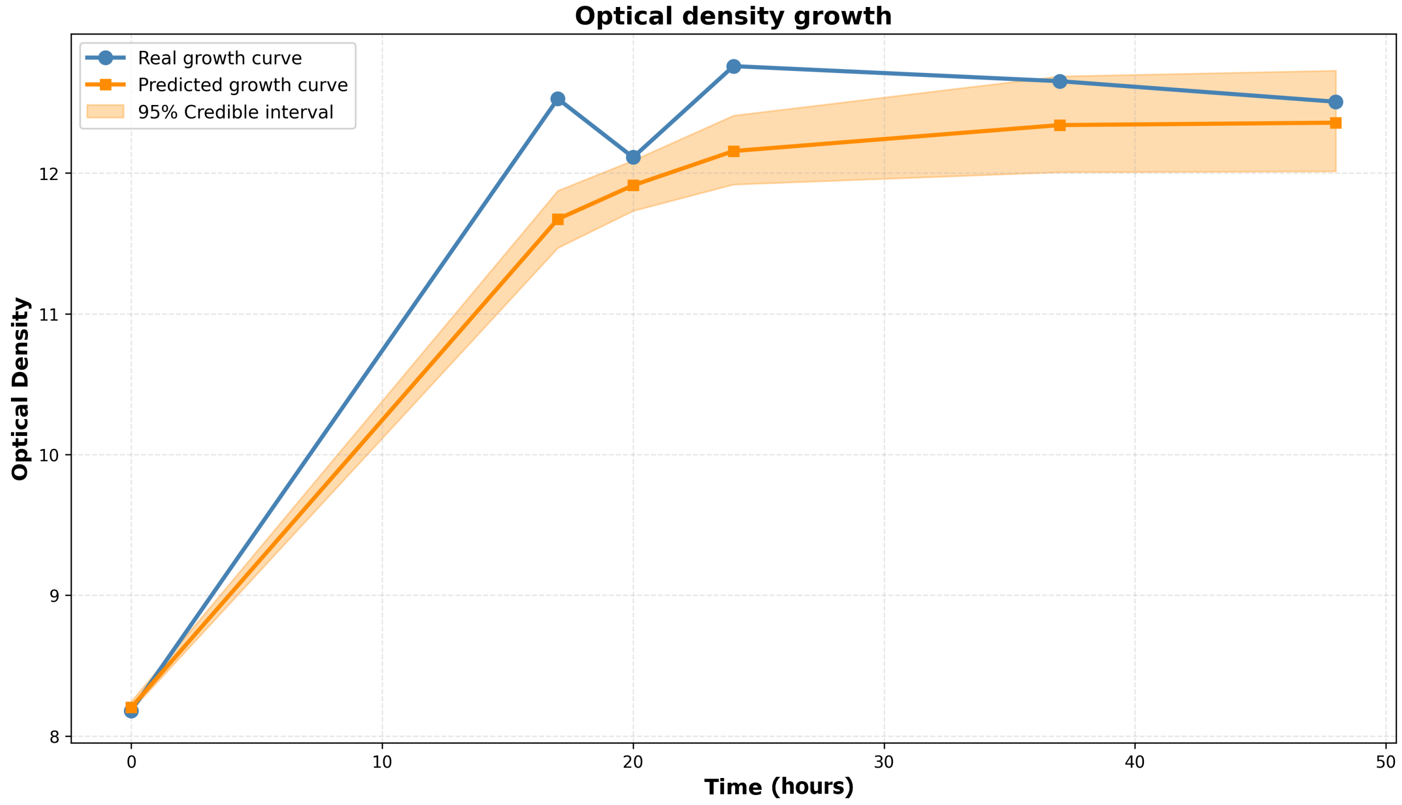

Error Distribution (K-Fold Cross-Validation)
Mean: 19.59%
Median: 10.48%
Density
0.020
0.015
0.010
0.005
0.000
0
50
100
150
200
Percentage Error (%)


Error Distribution (K-Fold Cross-Validation)
Mean: 6.59%
Median: 4.30%
Density
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
0
20
40
60
80
Percentage Error (%)

Predicted vs. Observed Growth Curve

Validation

The model demonstrates Feasibility for small-data bioprocess prediction: a solid foundation for a fully autonomous digital twin.



Actuation: Closing the Loop

The digital twin controls the **ultrasonic stimulation parameters**:

- **Frequency** — piezoelectric transducer driven at resonance via PWM square wave (ESP32 + H-bridge amplifier)
- **Duty cycle** — delivered as energy bursts modulated at 150 Hz
- **Temperature** — monitored; closed-loop thermal control is left for future work

> **Accelerating yeast growth** shortens the production cycle → significant economic benefit. The digital twin enables testing actuation strategies **without costly physical trials**.

Conclusions

- First **proof of concept** digital twin for an ultrasonic fermentation system
- Bayesian NN + Gompertz model **effective under data scarcity** (~100 points)
- Informative priors reduce MSE by **~7×** vs. non-informative baseline
- Real-time sensing pipeline (ESP32, piezoelectric disk, Dallas probe) **validated on physical fermenter**

⚖️ Core Message

The digital twin enables **faster, cheaper, smarter** fermentation optimisation — complementing slow physical experiments with rapid simulation and autonomous actuation.

Future Work

- **Expand dataset.** Collect more growth curves to relax prior constraints and improve generalisation
- **Beyond Gompertz.** Investigate alternative growth models to better capture yeast dynamics
- **Automate OD sensing.** Currently manual; real-time density measurement is the key bottleneck for full automation
- **Multiple MCMC chains.** Improve posterior exploration beyond the current single-chain implementation
- **Full closed-loop validation.** Experimentally validate the feedback-optimisation framework end-to-end

Most importantly, enjoy your beer.

Thank you!

Questions?

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Discussion

- How can **digital twin frameworks** best handle the tension between data scarcity and model accuracy in biological systems?
- What are the practical thresholds at which **Bayesian priors** can be relaxed in favour of data-driven learning?
- How should **ultrasonic actuation parameters** be explored efficiently — can the digital twin guide active learning?
- What does **intelligent environment** design look like for bio-oriented systems such as breweries?